PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program / Batch	B084C-Networkwalks
Date	13 November 2026
Modules Completed	**W2-PM1** (Footprinting with Multiple Kali Tools) **W2-PM2** (Footprinting & OSINT with Google Hacking Database - GHDB) **W2-PM5** (Network Scanning & Topology with Zenmap)
Client / Target	1. `networkwalks.com` (Authorized written permission secured) 2. Public Search Indexes via Exploit-DB GHDB (Passive OSINT) 3. Local LAN Subnet `10.57.180.0/24` (Owned local network environment)
Permission Secured from Client?	Yes (Authorization Ref: `NW-LOA-B084C-023`)
Phases Covered	**Phase 1:** Reconnaissance & Footprinting (Active CLI & Passive GHDB OSINT) **Phase 2:** Scanning & Network Discovery (Host Enumeration & Topology) **Phase 3–5:** In Progress

1. Liability Disclaimer

I have performed these activities only on the systems and devices where I had secured written permission or the devices and networks that I own myself. All these materials are for education and research purposes only. Do not use anything from here to break the law. The instructor, the authors, and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job, and a permanent record. In most countries, unauthorised access is a crime even when nothing is damaged.

2. Introduction

This consolidated report documents the Week 2 practical assessments completed as part of my Cybersecurity & Ethical Hacking internship at Networkwalks. It integrates active CLI footprinting against `networkwalks.com` using multiple Kali Linux tools (**W2-PM1**), passive OSINT reconnaissance using the Google Hacking Database (**W2-PM2**), and active local network discovery on subnet `10.57.180.0/24` using Zenmap (**W2-PM5**).

Together, these three modules demonstrate the progression from passive public intelligence gathering to active host enumeration and network topology mapping. Every step below documents the exact tools and commands executed, the observed technical outputs, security risk implications, and collected evidence.

3. Tools Used

The table below lists each platform, utility, and database used across Modules 1, 2, and 5 along with its specific reconnaissance or scanning purpose.

Tool / Utility	Module	Purpose
Kali Linux & Windows 10	All Modules	Hybrid operating environments used for CLI footprinting, OSINT queries, and GUI network scanning.
WHOIS	W2-PM1	Query domain registration records (registrar, registration/expiry dates, lock status, name servers).
WhatWeb	W2-PM1	Fingerprint web technologies including web server, CMS version, plugins, frameworks, and server IP.
nslookup	W2-PM1	Resolve the target domain name to its public IPv4 hosting address via DNS lookup.
curl -I	W2-PM1	Inspect HTTP response headers, server banners, cookies, security policies, and exposed API endpoints.
wafw00f	W2-PM1	Detect and fingerprint any Web Application Firewall (WAF) protecting the web application.
DNSRecon	W2-PM1	Enumerate DNS zone records including SOA, NS, MX, TXT/SPF policies, and cPanel SRV records.
Google Search & GHDB	W2-PM2	Execute Google Dorks from Exploit-DB GHDB to passively locate exposed IP cameras and open PDF directories.
Zenmap (Nmap 7.993 GUI)	W2-PM5	Sweep the local LAN subnet (`10.57.180.0/24`) to discover live hosts, IPs, MACs, and network topology.
Windows CMD (ipconfig)	W2-PM5	Identify local host IPv4 address, subnet mask, default gateway, and physical adapter MAC address.

4. Activities Performed

4.1 Footprinting & Reconnaissance with Multiple Kali Tools (W2-PM1)

I performed reconnaissance against `networkwalks.com` using six native Kali Linux CLI utilities—`whois`, `whatweb`, `nslookup`, `curl`, `wafw00f`, and `dnsrecon`—to map the domain's ownership, web application stack, perimeter defenses, and DNS infrastructure without launching intrusive attacks.

First, running `whois networkwalks.com` identified **GoDaddy.com, LLC** as the domain registrar, HostGator authoritative name servers (`ns6135.hostgator.com` and `ns6136.hostgator.com`), and active EPP registry transfer/deletion locks (`clientDeleteProhibited`, `clientTransferProhibited`). Next, `whatweb networkwalks.com` fingerprinted an **Apache** web server running **WordPress CMS** alongside **Bootstrap**, **jQuery 3.7.1**, and **WordPress Download Manager** hosted in the United States.

Using `nslookup networkwalks.com`, the domain resolved to public IPv4 address **192.232.216.135**. Inspecting HTTP response headers with `curl -I https://networkwalks.com` confirmed `HTTP/2 200` status, `server: Apache`, security headers (`referrer-policy`, `permissions-policy`), and exposed WordPress REST API links (`/wp-json/`). Running `wafw00f networkwalks.com` confirmed the site is protected behind a **ModSecurity (SpiderLabs)** Web Application Firewall. Finally, `dnsrecon -d networkwalks.com` enumerated mail exchange records (`mail.networkwalks.com`), SPF email authentication policies (`include:websitewelcome.com`), and cPanel autodiscovery `SRV` records.

4.2 Passive Footprinting & OSINT with Google Hacking Database (W2-PM2)

In the second module, I utilized the **Google Hacking Database (GHDB)** on Exploit-DB (`www.exploit-db.com`) and advanced Google search operators to perform zero-touch passive OSINT footprinting. Because search engines crawl and index misconfigured services, crafted search queries ("Google Dorks") surface exposed interfaces without sending direct traffic to target systems.

For **Task 1**, I queried GHDB under *Various Online Devices* and combined operators such as `intitle:"webcamXP" inurl:8080`, `intitle:"Live View / - AXIS"`, and `inurl:view/view.shtml` to locate and verify 10 publicly accessible IP camera feeds and document any unauthenticated panels. For **Task 2**, I executed open-directory search dorks including `intitle:index.of "parent directory" mathematics pdf` to identify 10 misconfigured web directories exposing downloadable mathematics PDF ebooks.

4.3 Network Scanning & Topology Discovery with Zenmap (W2-PM5)

For the active scanning module, I installed the official Nmap `7.993` Windows binary bundle (including Zenmap GUI and Npcap) on my Windows 10 host. Running `ipconfig` in Windows Command Prompt identified my active **Wireless LAN adapter Wi-Fi** configuration with IPv4 address **10.57.180.168**, subnet mask **255.255.255.0**, and default gateway **10.57.180.174**, defining the target scan range **10.57.180.0/24**.

In Zenmap, I entered target `10.57.180.0/24` and selected the **Ping scan** profile (`nmap -sn 10.57.180.0/24`) to sweep all 256 IPv4 addresses without intrusive port probing. The scan completed in **3.54 seconds** and discovered **3 live hosts** on my local network:

No.	Live IP Address	MAC Address	Latency	Vendor / Host Role
1	`10.57.180.21`	`6C:E8:C6:E5:F4:22`	0.026s	Earda Technologies (Client)
2	`10.57.180.174`	`4A:0F:0F:CD:EF:D8`	0.022s	Default Gateway / Hotspot
3	`10.57.180.168`	Local Wi-Fi Adapter (`ipconfig /all`)	Local	Local Scanning Windows PC

After verifying the host list, I opened the **Topology** tab in Zenmap, reviewed the legend indicators, and exported the radial network topology diagram in PDF format via **Save Graphic**.

5. Risk Analysis / Impact

Based on the technical intelligence gathered across Modules 1, 2, and 5, the following exposure observations and potential security impacts were identified:

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	**Web technology & CMS versions exposed**	WhatWeb identified WordPress, WP Download Manager, Bootstrap, and jQuery versions.	Attackers can cross-reference exposed software versions against CVE databases to plan targeted exploits.	• Medium
2	**Origin server IP address identifiable**	`nslookup` resolved `networkwalks.com` directly to `192.232.216.135` .	Exposes hosting infrastructure location and enables direct IP-level reconnaissance.	• Low
3	**HTTP headers & REST API route exposed**	`curl -I` returned `server: Apache` banner and `/wp-json/` endpoint link.	Assists stack fingerprinting and potential WordPress API metadata or user enumeration.	• Low
4	**WAF presence identifiable**	`wafw00f` detected ModSecurity (SpiderLabs) WAF in 2 requests.	Reveals perimeter filtering technology, prompting attackers to craft evasion payloads.	• Low
5	**DNS, mail & cPanel SRV records exposed**	DNSRecon enumerated NS, MX, SPF TXT, and cPanel autodiscovery SRV records.	Maps internal mail routing and hosting control panel endpoints for social engineering or phishing.	• Medium
6	**Unauthenticated IoT cameras & open directories indexed**	GHDB dorks surfaced 10 live IP camera panels and 10 open directory PDF listings.	Unauthorized surveillance access and unintended public leakage of internal files via search engines.	• High

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
7	**Multiple active hosts visible on local LAN**	Zenmap ping sweep discovered 3 live devices on subnet `10.57.180.0/24`.	Rogue or unauthorized devices connected to the local broadcast domain could intercept or pivot traffic.	● Medium

> **Risk Level Key:** ● High | ● Medium | ● Low
> **Assessment Scope Note:** The findings above represent passive reconnaissance and active host discovery observations rather than confirmed exploitable vulnerabilities. No intrusive exploitation or denial-of-service testing was performed during Week 2. Further authorized vulnerability validation would be required to confirm exploitability.

6. Recommendations

Based on the observations from these three modules, I recommend the following defensive controls and hardening measures:

- **Minimize Public Technology Fingerprints:** Suppress verbose server version banners (`ServerTokens Prod`, `ServerSignature Off`) and restrict unnecessary CMS/plugin metadata leaks.
- **Maintain Timely Patch Management:** Keep WordPress core, plugins (such as WP Download Manager), and underlying web server packages updated against current security advisories.
- **Harden HTTP Headers & API Access:** Audit HTTP response headers and restrict unauthenticated access to sensitive `/wp-json/` REST API endpoints where not publicly required.
- **Audit Public DNS Records:** Periodically review DNS zones, SPF/DMARC policies, and cPanel service records to ensure only essential records are exposed.
- **Maintain & Tune WAF Rules:** Keep ModSecurity (SpiderLabs) enabled in active blocking mode with updated OWASP Core Rule Sets and centralized alert logging.
- **Disable Web Directory Indexing & Secure IoT Devices:** Enforce `Options -Indexes` on web servers, restrict search engine indexing with `robots.txt` / `noindex`, and place IP cameras behind strong authentication and VPN segmentation.
- **Conduct Regular Internal Network Audits:** Perform scheduled LAN discovery scans to verify connected MAC/IP inventory, investigate unknown devices, and maintain up-to-date network topology documentation.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship at Networkwalks, I successfully executed structured reconnaissance and network scanning across three practical modules (**W2-PM1**, **W2-PM2**, and **W2-PM5**). Using Kali Linux CLI utilities, I mapped `networkwalks.com`'s registration, web stack, HTTP headers, ModSecurity WAF, and DNS architecture; using GHDB dorks, I demonstrated how search engines expose misconfigured cameras and open directories; and using Zenmap on Windows, I discovered all 3 live hosts on my `10.57.180.0/24` LAN subnet and generated its visual topology map.

These exercises reinforced that thorough information gathering forms the essential foundation of both offensive security assessments and proactive defense, and that all testing activities must remain strictly bounded by written client authorization.

8. Evidences Collected

All terminal command logs, dork verification tables, Zenmap scan outputs, and exported PDF topology diagrams have been archived in the internship repository. Key local network discovery and Zenmap execution evidences are presented below:

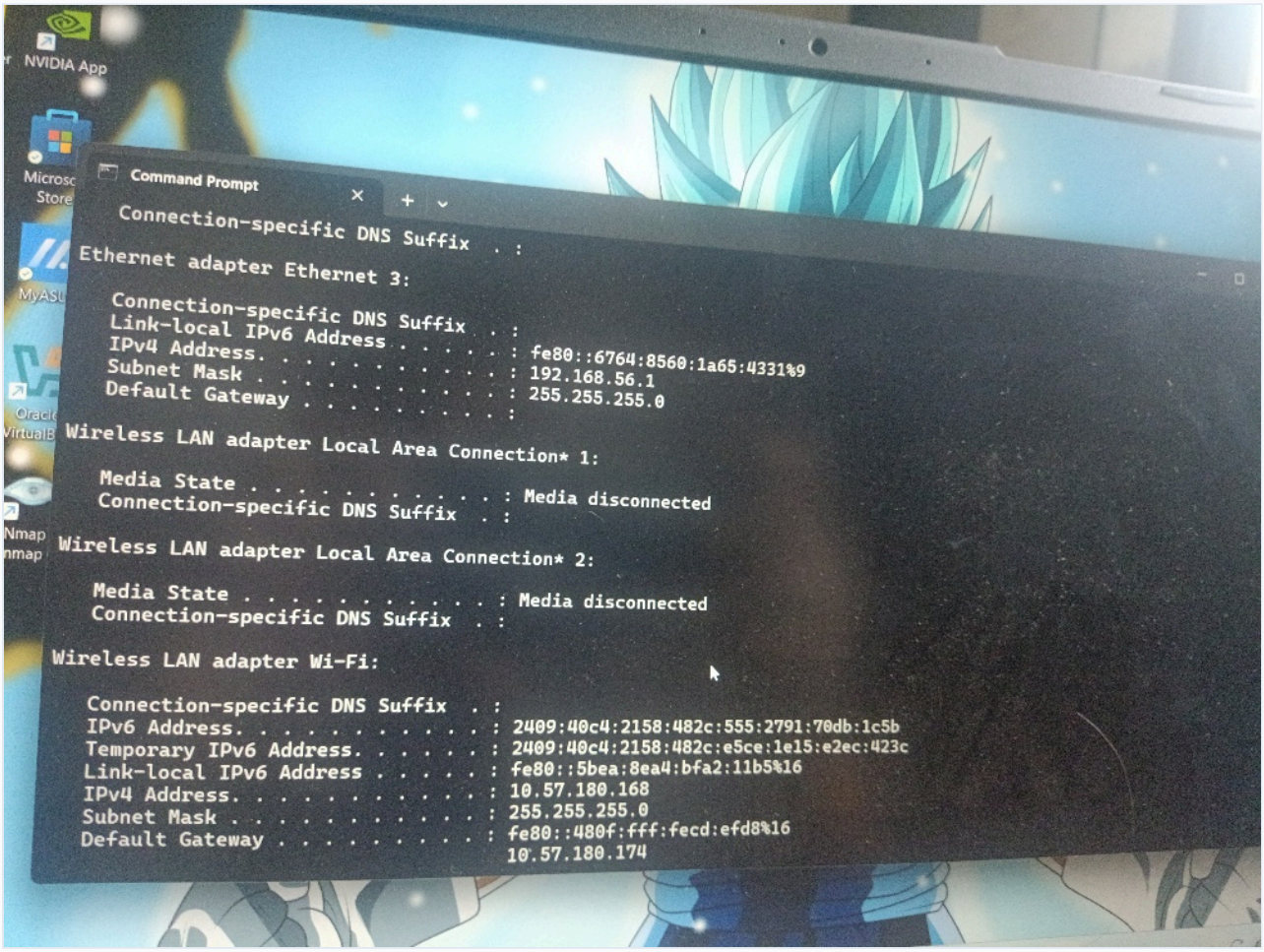


Figure 1 (W2-PM5 Task 2): Windows Command Prompt `ipconfig` output identifying active Wi-Fi IPv4 address `10.57.180.168`, subnet mask `255.255.255.0`, and gateway `10.57.180.174`.

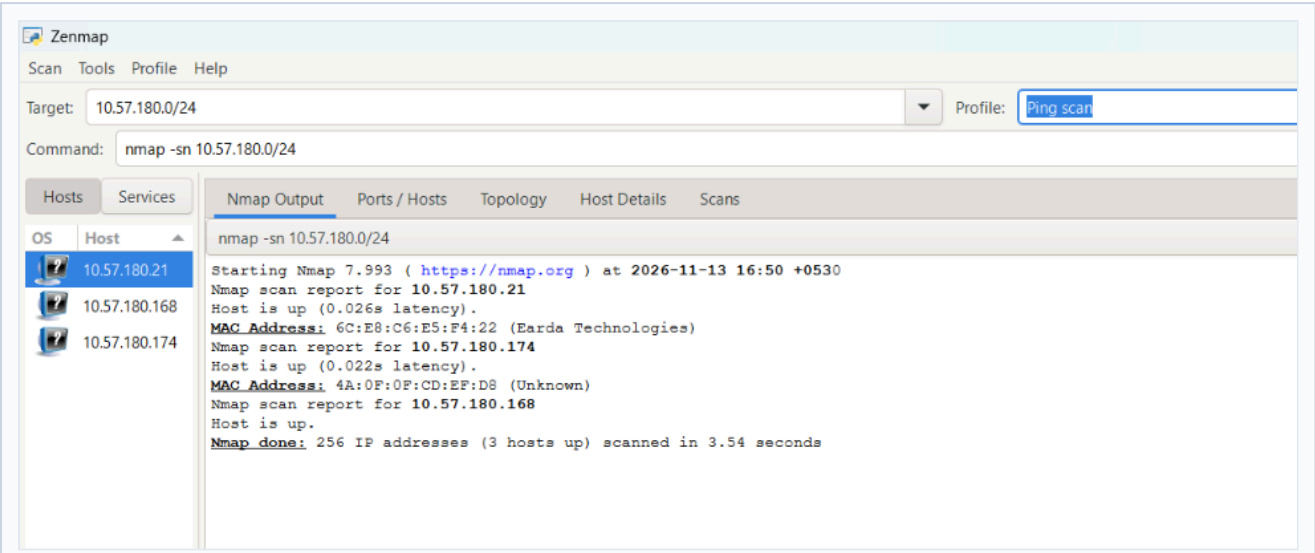


Figure 2 (W2-PM5 Tasks 3–6): Zenmap Ping Scan (`nmap -sn 10.57.180.0/24`) discovering 3 live hosts (`10.57.180.21` , `10.57.180.174` , and `10.57.180.168`) with MAC hardware identifiers in 3.54 seconds.

Module Ref	Archived Evidence Artifacts	Verification Status
W2-PM1	Kali Linux terminal screenshots & `.txt` outputs for `whois` , `whatweb` , `nslookup` , `curl -I` , `wafw00f` , and `dnsrecon` .	Verified & Archived
W2-PM2	Exploit-DB GHDB search captures and verified URL/dork tables for 10x live IP cameras and 10x mathematics PDF directories.	Verified & Archived
W2-PM5	Windows `ipconfig` output, Zenmap `nmap -sn 10.57.180.0/24` scan screenshot, and exported PDF Network Topology map.	Verified & Archived

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Project Information
Program Name: Cybersecurity & Ethical Hacking at Networkwalks
Week: 02 | Modules: W2-PM1, W2-PM2, W2-PM5 | Repository: GitHub